

Chapter Three:

Introduction to Artificial Intelligence (AI)

Introduction

- **Artificial Intelligence** is composed of two words, and each has its meaning.
 - ✓ **Artificial** defines as “Human-made”.
 - ✓ **Intelligence** defines as “Thinking power” or “The *ability to learn and solve problems*”.
- **Intelligence** is the ability to acquire and apply knowledge.
- **Knowledge** is the information acquired through *experience*.
- **Experience** is the knowledge gained through *exposure (training)*.
- **Intelligence** is composed of:
 - ✓ **Learning**: the *acquisition of knowledge or skills* through *study, experience, or being taught*.
 - ✓ **Reasoning**: the *act of thinking about something* in a *logical sensible* way.
 - ✓ **Problem-Solving**: the *process of finding solution to difficult or complex issues*.
 - ✓ **Perception**: the *ability to detect or become aware of something through the senses*.
 - ✓ **Linguistic Intelligence**: the *ability to understand and use spoken and written language*.

Learning



Reasoning



Problem



Solving

Perceiving



Understanding of language



Artificial Intelligence (AI)

- **Artificial Intelligence:** A human-made thinking power.
- **Artificial intelligence (AI)** is a wide-ranging branch of Computer Science concerned with building smart machines capable of performing tasks that typically require human intelligence.

Or

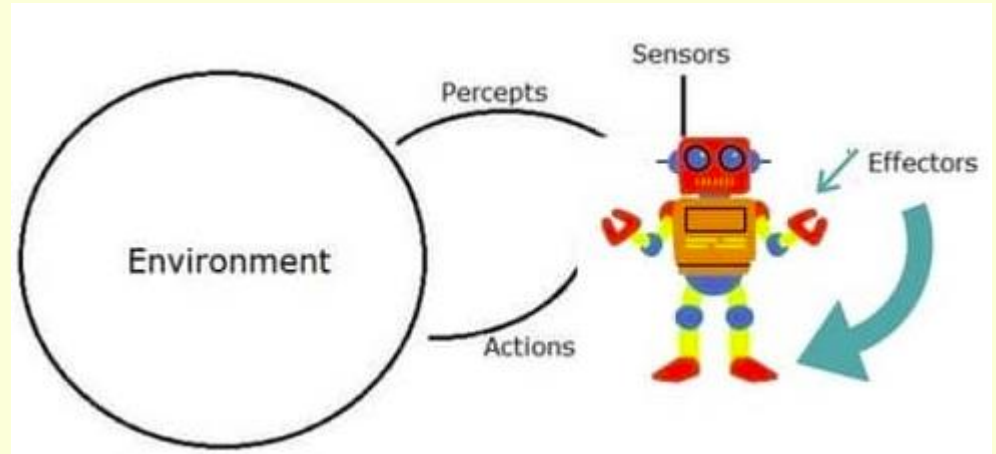
- Artificial Intelligence (AI) is the branch of computer science by which we can create intelligent machines which can behave like a human.
 - ✓ Think like humans, and able to make decisions.

- **Artificial Intelligence** *exists* when a machine can have human-based skills such as:
 - ✓ **Learning**,
 - ✓ **Reasoning**, and
 - ✓ **Problem solving**
- **Artificial Intelligence** machine *can perform* a task **without specifically being pre-programed** to do the task.

Agent and Environment

- An **AI system** is composed of an **agent** and its **environment**.
- **Agent**: is anything that can be viewed as:
 - ✓ Perceiving its environment through sensors, and
 - ✓ Acting upon that environment through actuators/effectors.
- **Examples of Agent**:
 - ✓ A Human-agent has eyes, ears, and other organs which act as sensors, and hands, legs, mouth, and other body parts acting as actuators.
 - ✓ A Robotic agent has cameras and infrared range finders which act as sensors and various motors acting as actuators/effectors.

- Intelligent agents must be able to set goals and achieve them.
- An **agent**:
 - ✓ Assess its environment,
 - ✓ Make predictions, and then
 - ✓ Evaluate its predictions and
 - ✓ **Adapt** based on its assessment



- **Machine perception** is the ability to **use input from sensors** (such as **cameras, microphones, sensors**, etc.) to **deduce aspects of the world**. e.g., Computer Vision
 - ✓ **Computer vision** is a field of *artificial intelligence (AI)* that **enables computers and systems** to **derive meaningful information** from *digital images, videos* and other **visual inputs** and **take actions** or *make recommendations* based on that information.
 - ✓ Computer vision describes machine understanding of images and videos.

High-Profile Examples of AI

- *Autonomous vehicles (such as drones and self-driving cars),*
- *Medical Diagnosis,*
- *Creating art (such as poetry),*
- *Proving mathematical theorems,*
- *Playing games (Alpha Go),*



- *Search engines (such as Google search),*
- *Online assistants (such as Siri),*
- *Image recognition in photographs,*
- *Spam filtering,*
- *Prediction of judicial decisions, and*
- *Targeting online advertisements.*

Enabling Technologies for the Advancement of AI

- *The arrival of the Internet,*
- *The advent of Big Data,*
- *Smart mobile phones,*
- *Social media,*
- *Cheaper and more powerful hardware:*
 - ✓ such as **Graphical Processing Units (GPUs)**

Relationships of AI, ML, and DL

- **AI** is an umbrella discipline that covers everything related to making machines smarter.
- **Machine Learning (ML)** is a subset of Artificial Intelligence.
- The **ML** systems can automatically learn and improve without explicitly being programmed and get *smarter* and *smarter* over time without *human intervention* .
- **ML** enables machines to learn and make predictions by recognizing patterns or depending on past experiences.
- **ML** **refers** to an *AI system that can self-learn* based on the *algorithm*.
- The *recommendation systems* on **music** and **video** streaming services are examples of ML.

- **Deep Learning (DL)** is a machine learning (ML) applied to large data sets.
- DL is inspired by the way a human brain filters information, and associated with learning from examples.
- DL systems help a computer model to filter the input data through layers to predict and classify information.
- DL processes information in the same manner as the human brain.
- It is used in technologies such as *driver-less cars*.

Need for Artificial Intelligence

1. To **Create Expert Systems** that exhibit **intelligent** behavior.
 - ✓ An expert system is an AI system simulating the decision-making ability of a human expert.
 - ❖ Expert is a person who is very knowledgeable about or skilful in a particular area.
 - ✓ Capability to learn, demonstrate, explain, and advice its users.
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- **Examples of Expert Systems:**

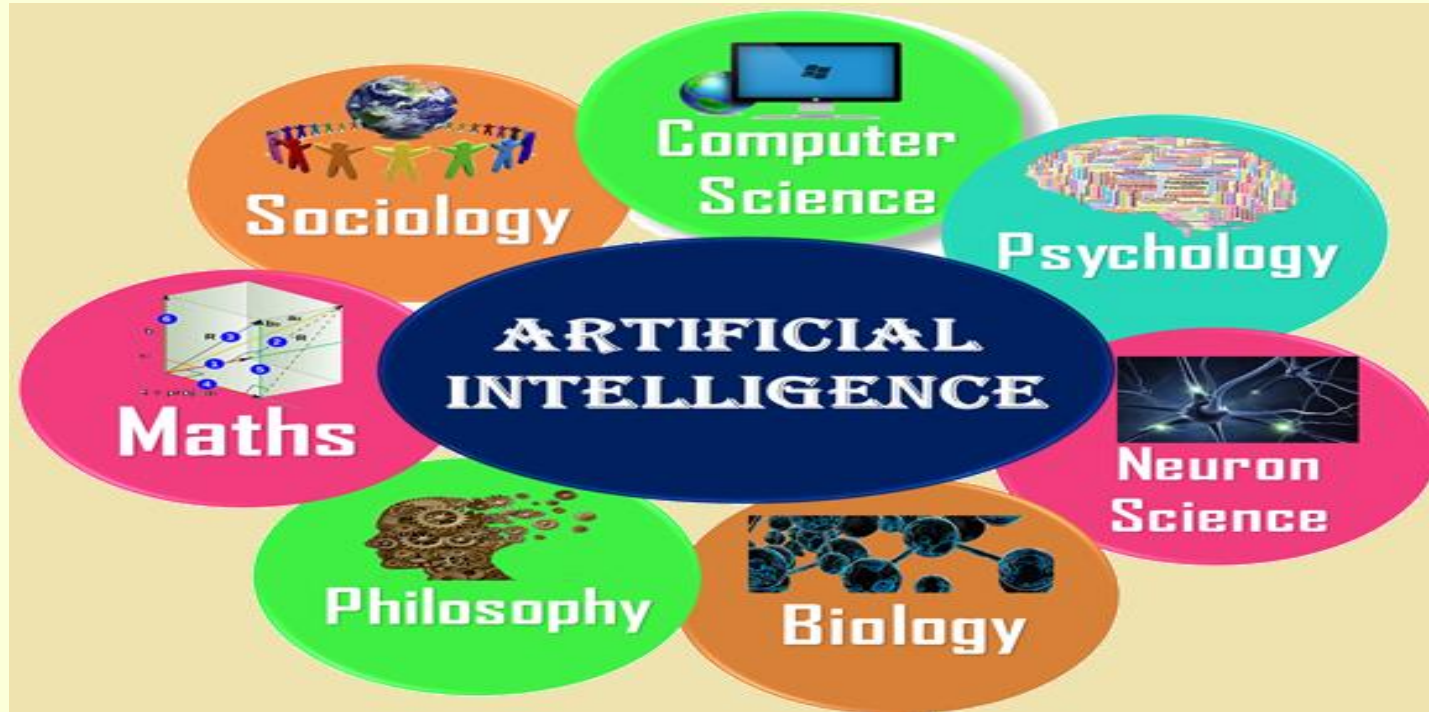
- ✓ **MYCIN:** is an expert system used to identify various bacteria that could cause severe infections.
 - ✓ It could also recommend drugs based on the patient's weight.
- ✓ **DENDRAL:** is an expert system used for chemical analysis to predict molecular structure.
- ✓ **PXDES:** an expert system used to predict the degree and type of lung cancer.
- ✓ **CaDet:** One of the best expert system that can identify cancer at early stages.

2. Helping machines **find solutions** to **complex problems** like **humans** do.

Goals of Artificial Intelligence

- Replicate human intelligence,
- Solve Knowledge-intensive tasks,
- An intelligent connection of perception and action,
- Build a machine which can perform tasks that requires human intelligence:
 - ✓ Plan some surgical operation,
 - ✓ Driving a car in traffic, and
 - ✓ Proving a theorem
- Creating some system which can exhibit intelligent behavior, learn new things by it self, demonstrate, and explain.

What Comprises to Artificial Intelligence



Advantages of Artificial Intelligence

- **High Accuracy with Fewer Errors:**
 - ✓ AI machines or systems are prone to fewer errors and high accuracy as it takes decisions as per pre-experience or information.
- **High-Speed:**
 - ✓ AI systems can be of very high-speed and fast-decision making, because of that AI systems can beat a chess champion in the Chess game.
- **High reliability:**
 - ✓ AI machines can perform the same action multiple times with high accuracy.

- **Useful for risky areas:**
 - ✓ AI machines can be helpful in situations such as defusing a bomb, exploring the ocean floor, where to employ a human can be risky.
- **Digital Assistant:**
 - ✓ AI can be very useful to provide digital assistant to users such as AI technology is currently used by various E-commerce websites to show the products as per customer requirements.
- **Useful as a public utility:**
 - ✓ Self-driving car which can make our journey safer and hassle-free, facial recognition for security purposes, Natural language processing (for search engines, for spelling checker, for assistant like Siri, for translation like Google translate), etc.

Disadvantages of Artificial Intelligence

- **High Cost:**
 - ✓ The hardware and software requirement of AI is very costly as it requires lots of maintenance to meet current world requirements.
- **Can't think out of the box:**
 - ✓ Limited to the work for which they are trained or programmed.
- **No feelings and emotions:**
 - ✓ AI machines does not have the feeling so it cannot make any kind of emotional attachment with humans.

- **Increase Dependence on Machines:**
 - ✓ People are getting more dependent on devices and hence they are losing their mental capabilities
- **No Original Creativity:**
 - ✓ As humans are so creative and can imagine some new ideas but still AI machines cannot beat this power of human intelligence and cannot be creative and imaginative.

Levels of Artificial Intelligence

1. Rule based AI Systems:

- These systems are simple artificial intelligence models which utilize the rule of if-then coding statements.
- Produces pre-defined outcomes that are based on a set of certain rules coded by humans.
- Rule-based systems require:
 - A set of facts or source of data, and
 - A set of rules for manipulating that data.
- The set of rules are sometimes referred to as '**If statements**' as they tend to follow the line of:
 - 'IF X happens THEN do Y'.
- A classic example of a rule-based system is the domain-specific **Expert System**.
 - ✓ An expert system might help a medical doctor choose the correct diagnosis based on a cluster of symptoms, or
 - ✓ Select tactical moves to play a game.

2. Context-Awareness and Retention:

- The machines are trained on the best human knowledge and experience with knowledge based on specific domains.
- Well known applications of this level are **chatbots** and “**roboadvisors**”.
 - ✓ A **Chatbot** is a computer program that simulates human conversation through voice commands or text chats or both.
 - ✓ **Chatbots** can understand a language, and you can speak with it more conversationally as if it's a person.
 - ✓ **Chatbots** can learn from each conversation with their users to get smarter over time.

→ <https://replika.ai/>

3. Domain Specific Expertise:

- Due to the capability to store and process a wide volume of information to make decisions on a specific domain the capabilities of these machines can be more than humans.
 - Cancer diagnosis is one of the applications in domain specific expertise.
 - Google Deepmind's system that defeated 18 times world go champion *Lee Sadol* proved the domain-specific expertise.

4. Reasoning Systems (Machines):

- This machine has a concept of intellect, intentions, knowledge, and their own logic.
- The machine has the capacity to reason, interact and, deal with other machines and humans too.
- It could reason or negotiate with humans and other machines.
- At the moment these algorithms are still in development.

5. Self-Aware Systems / Artificial General Intelligence (AGI):

- To develop a **machine** with **intelligence** like **human**.
- *However*, no such use is in it today.

6. Artificial Super Intelligence (ASI):

- The algorithms will be capable of defeating smartest of the humans in all the domains.
- This stage can be the real solution to the problems that are still complex for the human mind like poverty, hunger, and climate change.
- This will be the stage where machines made by humans will outsmart humans too.

7. Singularity and Transcendence/Excellency:

- The development path led by ASI will lead us to the point in time where the capabilities of humans will see tremendous extension.
- With the help of this ASI technology, humans would be able to connect the minds of each other and be like the human internet.
- The technology would be much like an imaginary life where ideas, and thoughts could be shared by just one flick of mind.
- Humans would be capable of connecting with other forms of life like animals, plants, and other natural activities. While some argue that humans will never reach such levels and some say that it is feasible and with this exponential growth rate would be achieved somewhere in 2045.

Types of Artificial Intelligence

- Mainly AI is categorized into two types, based on capabilities and functionalities :
 - ❖ **Type 1: Based on capabilities:**
 - ✓ Weak or Narrow AI,
 - ✓ Strong or General , and
 - ✓ Super AI
 - ❖ **Type 2: Based on functionality:**
 - ✓ Reactive Machines,
 - ✓ Limited Memory,
 - ✓ Theory of Mind, and
 - ✓ Self-Awareness

Type 1: Based on Capabilities:

1. Weak or Narrow AI:

- The most common and currently available AI in the world of Artificial Intelligence.
- It's limited to a specific or narrow area.
- It's able to perform a dedicated task with intelligence.
- Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task.

Examples:

- ✓ **Image and facial recognition systems:** These systems, including those used by social media companies like Facebook and Google to automatically identify people in photographs.
- ✓ **Self-driving vehicles:** Autonomous or semiautonomous cars, such as some Tesla models and autonomous drones, boats and factory robots, are all applications of narrow AI.
- ✓ **Google translate:** a free service offered by Google instantly translates words, phrases, and web pages between English and over 100 other languages.

2. Strong or General AI:

- Strong/General AI refers to AI that exhibits human-level intelligence.
- So, it can understand, think, and act the same way a human might in any given situation.
- In theory, then, anything a human can do, a strong or general AI can do too.
- General AI is a type of intelligence that could perform any intellectual task with efficiency like a human.
- The idea behind the general AI to make such a system that could be smarter and think like a human on its own.
- Currently, there is no such system exists which could come under general AI and can perform any task as perfect as a human.
- It may arrive within the next 20 or so years.

3. Super AI:

- It is an outcome of general AI.
- Super AI refers to an intelligent machine that surpasses human intelligence and ability.
- This refers to aspects like general wisdom, problem-solving and creativity.
- It's also known as *artificial superintelligence (ASI)* or *superintelligence*.
- It's still a hypothetical concept of Artificial Intelligence.

Type 2: Based on Functionality:

1. Reactive Artificial Intelligence:

- It's the simplest level of AI.
- React to some input with some output.
- There is no learning that occurs, and this is the first stage to any A.I. system.
- Do not store memories or past experiences for future actions.
- Reactive AI functions the way it was programmed with a predictable output based on the input it receives.
- These machines only focus on current scenarios and react on it as per possible best action.
- **IBM's Deep Blue system** is an example of *reactive machines*.
- Google's **AlphaGo** is also an example of reactive machines.

2. **Limited Memory Artificial Intelligence:**

- Limited memory machines can store past experiences or some data for a short period of time.
- These machines can use stored data for a limited time period only.
- Self-driving cars are one of the best examples of Limited Memory systems. Self-driving cars can store:
 - ✓ The recent speed of nearby cars,
 - ✓ The distance of other cars, speed limits, and
 - ✓ Other information to navigate the road.

3. Theory of Mind Artificial Intelligence:

- When machines acquire decision-making capabilities equal to humans, we will have achieved theory of mind AI.
- Theory of Mind AI should understand human emotions, people, beliefs, and be able to interact socially like humans.
- This type of AI machines is still not developed, but researchers are making lots of efforts and improvement for developing such AI machines.

4. Self-Aware Artificial Intelligence (artificial superintelligence):

- Self-awareness AI is the future of Artificial Intelligence.
- These machines will be super intelligent and will have their own consciousness, sentiments, and self-awareness.
- This type of AI is not only aware of the mental state of other entities but is also aware of itself.
- *Self-aware AI, or artificial superintelligence (ASI)*, is defined as a machine with intelligence on par with human general intelligence and in principle capable of far surpassing human cognition by creating ever more intelligent versions of itself.
- These machines will be smarter than the human mind.
- Self-Awareness AI does not exist in reality still and it is a hypothetical concept.

How Humans Think

- Human Intelligence or the cognitive process is composed of three main stages:
 - ✓ Observe and input the information or data in the brain.
 - ✓ Interpret and evaluate the input that is received from the surrounding environment.
 - ✓ Make decisions as a reaction towards what you received as input and interpreted and evaluated.

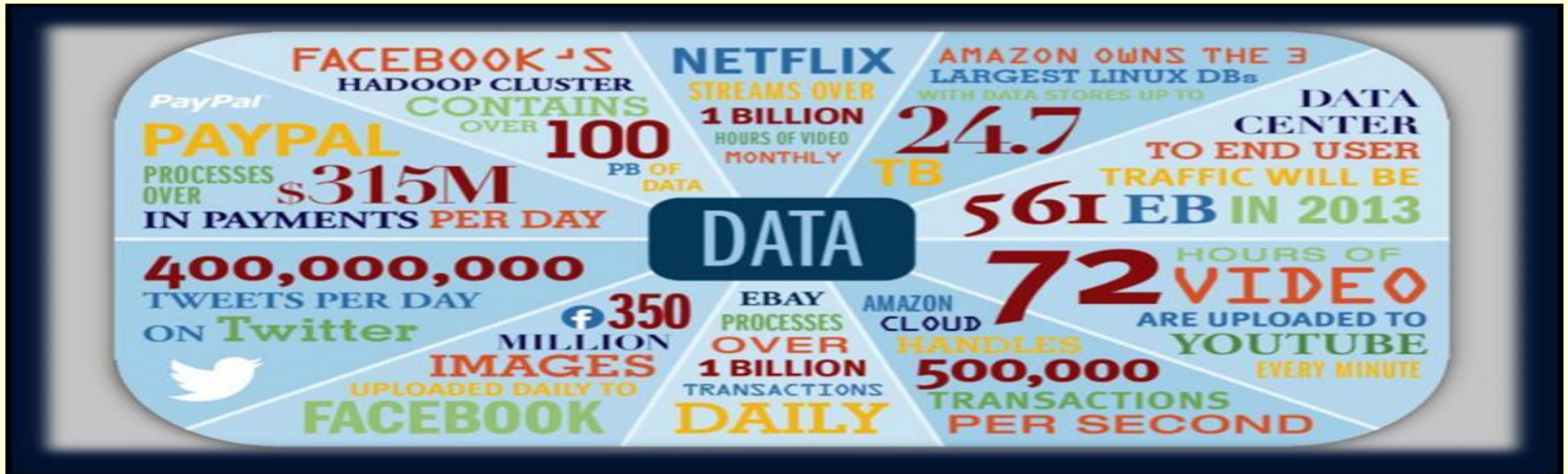
Mapping Human Thinking to AI Components

- Because AI is the study of mimicking human thinking, the phases of human thinking can be mapped to the layers or components of AI systems.
- In the first stage, humans acquire information from their surrounding environments through human senses, such as sight, hearing, smell, taste, and touch, through human organs, such as eyes, ears, and other sensing organs, for example, the hands.
 - ✓ In AI models, this stage is represented by the sensing layer, which perceives information from the surrounding environment.
 - ✓ This information is specific to the AI application.
 - ✓ For example, there are sensing agents such as voice recognition for sensing voice and visual imaging recognition for sensing images. Thus, these agents or sensors take the role of the hearing and sight senses in humans.

- The second stage is related to interpreting and evaluating the input data.
 - ✓ In AI, this stage is represented by the interpretation layer, that is, reasoning and thinking about the gathered input that is acquired by the sensing layer.
- The third stage is related to taking action or making decisions.
 - ✓ After evaluating the input data, the interacting layer performs the necessary tasks.
 - ✓ Robotic movement control and speech generation are examples of functions that are implemented in the interacting layer.

Influencers of Artificial Intelligence

- **Big Data**
- Advancements in computer **processing speed** and **new chip architectures**.
- **Enabled processing** of a large amount of data at **high speed**.



- **Cloud computing** and **APIs**:
 - ✓ **Cloud computing** made possible the availability of services such as *data analysis*, *social media*, *video storage*, *e-commerce*, and *AI capabilities* through the *Internet*.
 - ✓ **APIs** enable software components to communicate with each other easily.
- The **emergence** of **Data science**:
 - ✓ Data science uses machine learning and AI to process big data.

Example Services available using APIs and Cloud infrastructure:

- ✓ **IBM** delivers **Watson AI** services over **IBM Cloud**.
- ✓ **Amazon AI** services are delivered over **Amazon Web Services (AWS)**.
- ✓ **Microsoft AI** tools are available over the **MS Azure cloud**.
- ✓ **Google AI** services are available in the **Google Cloud Platform**.

Applications of Artificial Intelligence

- In **Agriculture**:
 - ✓ Agriculture robotics, soil and crop monitoring, predictive analysis.
- In **Healthcare**:
 - ✓ To make a better and faster diagnosis than humans.
- In **Education**:
 - ✓ AI chatbot can communicate with students as a teaching assistant.

- In **Finance and E-commerce:**
 - ✓ Adaptive intelligence, algorithm trading, help shoppers discover products.
- In **Gaming and Entertainment:**
 - ✓ Services such as Netflix show AI backed recommendations for programs or shows.
- In **Data Security:**
 - ✓ To determine software bugs and cyber-attacks

- In **Social Media:**

- ✓ Analyze lots of data to identify the latest trends, hashtags, and requirements of different users.

- In **Travel & Transport:**

- Making travel arrangements, suggesting hotels, flights, and best routes

- In **Automotive Industry:**

- ✓ Self-driven cars which can make your journey more safe and secure.

- In **Robotics:**

- To create intelligent robots which can perform tasks with their own experiences without pre-programmed.

Artificial Intelligence Tools and Platforms

- AI has developed a large number of tools to solve the most difficult problems in computer science such as:
 - ✓ **Search** and **optimization**: a process of making a website better for search engines.
 - ✓ **Logic**: computability theory.
 - ✓ **Probabilistic** methods for uncertain reasoning.
 - ✓ **Classifiers** and **statistical learning methods**.
 - ✓ **Neural networks**: a series of algorithms that mimic the operations of a human brain to recognize relationships between vast amounts of data.
 - ✓ **Control theory**: is the use of feedback to influence the behavior of a system to achieve the desired goal.
 - ✓ **Languages**: automatic manipulation of natural language, like speech and text, by software.

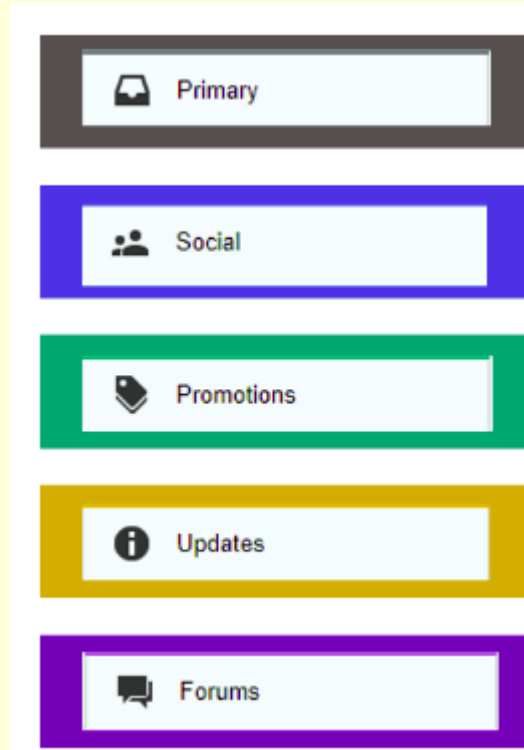
- Most common artificial intelligence platforms:
 - ✓ Microsoft Azure Machine Learning
 - ✓ Google Cloud Prediction API
 - ✓ IBM Watson
 - ✓ TensorFlow

Sample AI Applications

- **Commuting:**
 - ✓ Google's AI-Powered Predictions
 - ✓ Ridesharing Apps Like Uber and Lyft
 - ✓ Commercial Flights Use an AI Autopilot



- **Email:**
 - Spam Filters
 - Smart Email Categorization



- **Social Networking:**
 - Automatically highlighting faces and suggests friends tag, and
 - Add animated effects or digital masks.



- **Online Shopping:**

- ✓ You see recommendations for products you're interested in as “customers who viewed this item also viewed” and “customers who bought this item also bought”

Customer recommendations for MSI laptops on euro.message.

Customers Who Viewed This Item Also Viewed

Product Details:

- MSI Laptop (17 inch)
- Fiyat : 2,999.00 USD + KDV (İndirim Oranı % 17)
- İndirilmiş : 2,499.07 USD + KDV
- KDV Dahil : 5,243.14 TL | 12 x 436.93 TL taksitle
- Havale / EFT : 5,138.28 TL (KDV Dahil)

SEPETE EKLE

★★★★★ (0) Yorum oku | Yorum yap

Hızlı Gözetim | Kredi Ödeme

İlginizi Çekebilecek Diğer Ürünler

Product	Price	Reviews
ASUS C75VX-CV056H i7-3630QM 2.4GHz 16GB 750GB	4.399,20 TL	1 Yorum ★★★★★
MSI CE60 GND-412TR i7-3630QM 2.4GHz 8GB 750GB 2GB	1.353,83 USD	5 Yorum ★★★★★
ASUS N56VZ-S4283H i7-3630QM 2.40GHz 16GB 1TB 4GB Nvidia	2.748,91 TL	26 Yorum ★★★★★
TOSHIBA Qosmio X670-13C-3D i7-3630QM 2.4GHz 16GB 1TB	4.906,85 TL	0 Yorum ★★★★★
DELL Inspiron 3521 831W67C Intel Core i5-3317U 1.7GHz 6GB	1.298,91 TL	1 Yorum ★★★★★

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- **Mobile Use:**

- ✓ **Voice-to-Text:**

“Ok Google” for example.

- ✓ **Smart Personal Assistants:**

perform internet searches, set reminders, and calendar using Bots

